

. Quiz1 2026
Biophysics(BIO215)

Total Marks:20

Time: 45 minutes

Section-A (MCQ 1*10 Marks)

1. A student analyzes an X-ray diffraction image of an unknown biological polymer and observes a strong X-shaped pattern similar to Rosalind Franklin's experiment. What molecule is most consistent with this diffraction evidence?
A. The sequence of nucleotides
B. Presence of ribose sugar
C. The helical structure
D. Nitrogenous bases
2. A researcher mutates a residue at a tight turn in a protein to glycine. The protein becomes more flexible but loses stability. Which explanation best accounts for this?
A. Glycine increases steric hindrance
B. Glycine lacks a side chain, increasing conformational entropy
C. Glycine forms fewer hydrogen bonds
D. Glycine disrupts peptide bond resonance
3. Scientists rejected Pauling's triple-helix DNA model mainly because:
A. The central arrangement would prevent stable hydrogen bonding between bases
B. The phosphate groups were packed in the core, creating repelling electrostatic charges
C. The model predicted excessive helical flexibility incompatible with genetic storage
D. The sugar backbone could not form covalent bonds at physiological pH
4. A protein is expected to produce an unusually complex X-ray diffraction pattern because of its large number of atoms and subunits. Which listed protein best fits this expectation?
A. RNA polymerase
B. Hexokinase
C. Chymotrypsin
D. Myoglobin
5. Hemoglobin required years of X-ray analysis because protein crystals scatter X-rays in highly complex patterns. What does this complexity primarily reflect?
A. Protein is structurally uniform
B. Proteins contain many atoms arranged in a tertiary structure
C. Protein lacks symmetrical structure
D. Protein contains repeating units that scatter x-rays in simple pattern
6. If a base X' pairs both with A' and T' but A'–T' pairing is twice the frequency of each individual X' pairing, and the G' base fraction is 15% (bases), what is the %C' (bases)?
A. 15%
B. 17.5%

- C. 23.33%
- D. 26.25%

7. While modeling a protein backbone, a researcher notices that most peptide bonds naturally settle into one preferred geometry without atomic crowding. This geometry allows adjacent residues to remain spaced apart and stable.

What conformation is being observed:

- A. Trans conformation
- B. Cis conformation
- C. Bend conformation
- D. Random coil conformation

8. A student enters a command while visualizing protein structure in chimera: `disp:220 z<8` What is the goal of this command?

- A. Display residue 220 only
- B. Display residues farther than 8 Å
- C. Display residues within 8 Å of residue 220
- D. Zoom into residue 220

9. When DNA is placed in water, the bases remain tightly packed inside the helix, while the sugar-phosphate backbone stays exposed to the surrounding solution. Which structural property best explains this arrangement?

- A. The core is filled with water molecules to facilitate hydrogen bonding.
- B. Both the core and the exterior are equally hydrophilic.
- C. The core is hydrophobic while the exterior is hydrophilic.
- D. The core is hydrophilic, while the exterior is hydrophobic.

10. During structural analysis in chimera, a researcher finds a protein segment tightly packed with nonpolar residues and consistently excluded from contact with surrounding water molecules. This segment most likely evolved to function in which environment?

- A. Cytoplasmic solvent exposure
- B. Lipid membrane interior
- C. Active catalytic site
- D. Nuclear pore passage

Section-B (2*5=10 mark)

1. Justify replacing the Buckingham potential with the Lennard-Jones model in molecular dynamics.

Ans. Ans: The Buckingham potential uses an exponential term ($A\exp^{-B/r}$) to model quantum repulsion. Calculating exponential functions is computationally expensive and slow for a computer, especially when done for millions of atom pairs in a simulation. [1 mark] Computational Efficiency: The Lennard-Jones potential uses a $1/r^{12}$ term for repulsion. This is extremely efficient to compute because the attractive $1/r^6$ term must already be calculated for the dispersion force. The repulsive term can be found with just one additional multiplication: $(1/r^6) * (1/r^6)$. [1 mark]

2. A protein is unfolded in 8 M urea and then the urea is removed. The protein refolds to its original active form within seconds. Explain how this observation supports Anfinsen's hypothesis but appears to contradict Levinthal's paradox.

Support for Anfinsen: The spontaneous refolding proves that the primary amino acid sequence contains all the necessary information to reach the most thermodynamically stable native state without external templates. [1 mark] Contradiction of Levinthal: The "seconds" timescale proves folding cannot be a random search of all possible conformations (which would take eons); instead, it must follow a directed, energy-minimized pathway or funnel. [1 mark]

3. A researcher studies two proteins of similar length: one functions as a globular enzyme while the other serves a structural role. Explain why only one exhibits catalytic activity and describe the structural differences that distinguish them.(2 marks)

The catalytic protein is globular and folds into a complex 3D structure [0.5] This folding creates a specific active site required for catalysis [0.5] The structural protein is fibrous with a repetitive, extended arrangement [0.5] Fibrous proteins are optimized for strength rather than chemical activity [0.5]

4. Why do proteins lose their native structure when exposed to high temperature, even though peptide bonds remain intact?

Ans: Proteins lose their native structure at high temperature because heat disrupts non-bonded interactions such as hydrogen bonds, ionic interactions, and hydrophobic interactions that stabilize the secondary and tertiary structure. The peptide bonds remain intact, but the loss of these weak forces causes denaturation of the protein.

5. In a DNA molecule, the ratio of $(A + T)/(G + C) = 1.5$. If total base pairs = 2000,
 - a. calculate the number of AT and GC base pairs.
 - b. Total number of phosphodiester bonds.
 - a. $(A + T)/(G + C) = 1.5$, $A+T=x$ & $G+C=y$, $x/y=1.5=3/2$

$$x:y=3:2 \quad AT=1200 \text{ and } GC= 800$$

- b. Each strand with n nucleotides has $n-1$ phosphodiester bonds, So bonds per strand $=2000-1 = 1999$

$$\text{Total number of phosphodiester bonds} = 2 \times 1999 = 3998$$